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Roll No

EE-8004(2) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Power System Planning and Reliability
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the basic concept of Probability theory. What are the rules for combining probabilities of events?
b) What is Binomial and Poisson distributions? Explain expected value and standard deviation for continuous and discrete distributions.
2. a) Define Minimal cut-set. Explain the reliability analysis of complex systems using minimal cut-set method.
b) Differentiate in short term, medium term and long term planning.
3. a) Explain two static Markov model and derive the expression of availability and unavailability.
b) What are the impact of weather on transmission lines? Explain by model.
4. a) What are the different component reliability? Explain Hazard rate.
b) A system has a constant failure rate of 0.001 failures per hour. What is the probability that this system will fail before $t = 1000$ hours? Determine the probability that it will work for at least 1000 hours.
5. a) What are the various major steps involve in Operating reserve calculations?
b) Discuss the effect of varying load on composite system reliability.
6. a) Explain the correlated and uncorrelated load models.
b) Explain the load modeling with cumulative load model.
7. a) Discuss the reliability of a system with non-repairable components.
b) Explain in detail the different techniques employed in improving the reliability.
8. Write short note on (any two)
 - a) Cost and Worth of reliability
 - b) Distribution system reliability
 - c) Weibull distribution
